Causal Regime-Conditioned Trading Agent: Verifiable Regime-Aware Reinforcement Learning

Varad Durge, Ishaan Kawathekar, Tejas Putta
Dept. of Electronics & Computer Engineering
SCTR's Pune Institute of Computer Technology
Pune, India
varaddurge@gmail.com [TODO: verify co-author e-mails]

Dr. Anjali A. Yadav
Dept. of Electronics & Computer Engineering
SCTR's Pune Institute of Computer Technology
Pune, India
[TODO: verify guide e-mail and author order]

***Abstract*—Regime-aware reinforcement-learning (RL) trading agents are usually judged by back-tested returns, which cannot show whether the regime signal actually drives their decisions. We present a reproducible pipeline that couples a Wasserstein regime-identity tracker with a suite of interventional tests on the trained policy. A four-state Gaussian hidden Markov model (HMM) is refitted every 42 four-hour bars on past rows only. Each new component is matched to persistent bull, bear, calm and crisis templates by the closed-form 2-Wasserstein distance, so regime names survive refits instead of being re-sorted. Two proximal policy optimisation (PPO) agents, identical except for four regime-probability inputs, are trained walk-forward on BTC/USDT data from 2021 to 2026. Every out-of-fold decision is audited by input ablation, counterfactual regime injection and a crisis-probability sweep. On 11,430 test bars, the tracker lowers the mean template-matching cost by a factor of 4.9 and gives labels whose volatility ordering is stable, but its non-injective matching starves the bull template, which is never issued after burn-in. The conditioned agent clearly uses its regime input: ablation moves its position on 60.7% of bars, and it de-risks in bear and crisis segments. Yet only 0.1% of bars pass all three checks, against 33.5% for an archived run with moment-sorted labels. Reference policies confirm that the suite separates a faithful rule (97.9%) from ignoring and inverted rules (0%). The agent's Sharpe ratio of -0.16 (95% CI $[-0.98, 0.63]$) beats its blind twin (-1.12) but not buy-and-hold (0.35). A live paper-trading console runs both twins on Binance candles as they close. For 15-min and 1-h candles we train timeframe-native twins on BTC/USDT and ETH/USDT from 2021 with random 2,000-bar episodes, after the single-episode recipe collapsed to buy-and-hold on 1-h data. The recipe passed validation, yet on the holdout year five of the eight native twins again stay at or near a fixed position. Only the 1-h ETH conditioned agent trades on its regime input (ablation 27.2%), gaining 62.6% against -34.1% for its blind twin, with the gap not significant ($P(\text{gap}{>}0) = 0.91$). Over the three weeks after the study (22 Sep–13 Oct 2026), the blind twin finished ahead on 15-min, 1-h and 4-h BTC candles ($+3.4$, $+5.7$ and $+6.2$% against -4.1, -3.3 and -4.8%) and on 15-min ETH candles, and the twins finished level on 1-h ETH. No holdout or live gap between the twins is significant under a block bootstrap. Faithfulness must therefore be re-verified whenever the regime layer or the training recipe changes.**

***Index Terms*—regime detection, hidden Markov model, Wasserstein distance, optimal transport, reinforcement learning, algorithmic trading, explainability, faithfulness, counterfactual testing**

I. Introduction

Cryptocurrency markets move through very different conditions within a few years. Bitcoin rose sharply in 2021, lost roughly three quarters of its value between late 2021 and late 2022, and then spent 2023–2026 alternating between recoveries and drawdowns. A trading policy fitted to such a history without any notion of market state tends to learn one averaged behaviour that suits none of these phases. A natural remedy is to detect the prevailing regime and hand it to the policy as an extra input. Ndoutoumou *et al.* combine a hidden Markov model with reinforcement-learning portfolio agents and report stronger risk-adjusted returns than the same agents without the regime signal [1]. RegimeFolio builds sector portfolios around detected regimes in non-stationary markets [2], and explainable classifiers have been used to allocate assets by market state [3].

What this work rarely checks is whether the regime input matters. A regime-conditioned agent can report that it "cut exposure because a crisis was detected" while its actual computation ignores the crisis probability, or responds to it in the wrong direction. An explanation that is easy to understand is not necessarily faithful to the computation that produced the decision [4]. Reviews of explainable machine learning find that explanations are often judged by whether users trust them, while standard tests of whether an explanation is true of the model are still missing [5]. Finance relies on much the same post-hoc tools, and their evaluation is uneven [6]. In high-stakes settings, the duty to explain a decision is not optional [7].

A second, quieter problem sits inside the regime detector. An HMM refitted on new data returns its states in an arbitrary order, so "state 2" may mean crisis in one window and calm in the next. Sorting states by their moments after every fit repairs the names locally, but it forces four names onto whatever four components the latest fit produces, even if one of them no longer resembles anything seen before. We instead keep persistent bull, bear, calm and crisis templates and match each new component to the nearest template by the closed-form 2-Wasserstein distance, so a name survives a refit only when the component still resembles it.

This paper asks two linked questions. Does Wasserstein

tracking give regime labels that keep their meaning across walk-forward refits? And when an RL agent is conditioned on those labels, does its behaviour depend on them in the way the labels' names imply? We answer both with a controlled experiment on BTC/USDT four-hour bars. Our contributions are:

- A walk-forward implementation of Wasserstein template tracking for a four-state Gaussian HMM, with the closed-form diagonal W_2 distance, probability aggregation and template smoothing written out as equations that match the released code.
- A per-decision verification suite of three interventions on the regime input (ablation, counterfactual injection and a sensitivity sweep), validated on reference policies that are faithful, regime-ignoring and inverted by construction.
- A twin experiment in which regime-conditioned and regime-blind PPO agents share architecture, seed and training budget, evaluated walk-forward with bootstrapped confidence intervals and a per-regime breakdown.
- An empirical finding: replacing moment sorting with Wasserstein tracking leaves the regime input load-bearing but collapses the faithfulness rate from 33.5% to 0.1%. We trace this to template starvation and to counterfactual inputs that lie outside the training distribution.
- A live paper-trading console that runs both twins on Binance candles as they close, on 15-min, 1-h and 4-h timeframes, and a frozen live study that compares their returns on candles closed after the research data, with a paired block bootstrap of the return gap. On 15-min and 1-h candles the twins are trained on their own timeframe, for BTC/USDT and ETH/USDT, with a validation, holdout and live split. We report how often this recipe still collapses to a fixed position.

II. Related Work

A. Regime detection

Ndoutoumou, Yin and Cheng fit a Gaussian HMM and pass the inferred regime into reinforcement-learning agents on a Dow Jones portfolio. For several algorithms the hybrid agent improves the Sharpe and Sortino ratios relative to the same agent without the HMM [1]. The comparison is on back-tested returns. It does not test whether the learned policy actually depends on the regime input. Tsang and Chen detect regime change in the foreign-exchange market with directional-change indicators, and use the detector to chart the period around Brexit [8]. Zhang, Yi and Chen use explainable machine learning to classify regimes and to support asset allocation [3]. RegimeFolio is a regime-aware learning system for sectoral portfolio optimisation when markets are non-stationary [2]. Standard HMMs switch state too readily on noisy data; we suppress this with a simple minimum-dwell rule, and we keep regime names stable across refits by Wasserstein template matching, described in Section IV.

B. Reinforcement learning for portfolio optimisation

Deep reinforcement learning is now a standard way to treat portfolio choice as a sequential decision. Surveys cover the single-agent literature and the move toward multi-agent systems [9], [10]. Wang, Jin and Li cast asset-weight management as a Markov decision process and train a deep RL agent for diversification [11]. Tamuly, Bhutani and Sukriti apply deep RL to dynamic capital allocation [12]. Naik *et al.* use proximal policy optimisation and deep deterministic policy gradient for real-time rebalancing on Indian equities, with the Sharpe ratio as the risk-adjusted objective [13]. Huang and Jing adapt a single-stock portfolio with a deep deterministic policy gradient agent [14]. Gao and Tang evaluate advantage actor-critic and proximal policy optimisation on A-share portfolios drawn from CSI 300 constituents, under A-share trading constraints [15]. Shrirao *et al.* train portfolio policies intended to remain usable under adversarial market conditions [16]. Across this work, a higher Sharpe ratio is treated as evidence that the agent has adapted. None of it checks whether a named regime input is what the policy uses.

C. Explainability

Mamalakis *et al.* separate an explanation that is faithful to a deep network from one that is merely comprehensible [4]. Wiratsin and Ragkhitwetsagul review whether explainable-AI techniques improve human trust, and they note the lack of standard evaluation metrics [5]. Reinhardt and Buchholz argue that explainability in high-stakes domains is an obligation, not only a convenience [7]. Yeo *et al.* survey how post-hoc explanations are used in finance and how their limits affect decisions [6]. Our tests are different in kind: they intervene on one named input of a sequential policy rather than attributing a static prediction.

D. Causal inference in financial decisions

Qiu *et al.* build a causal-inference model for stock-trend forecasting, on the view that predictive accuracy alone does not identify which dependencies the forecast uses [17]. Katamaneni *et al.* study machine-learning causal inference for decisions that must be revised as the environment changes [18]. Both treat causality as a property of the market time series. The interventions in this paper are different. The policy is a known function, so setting its regime input to a chosen value measures the effect of that input on the action, not a causal effect in the market.

III. Preliminaries

A. Gaussian hidden Markov model

Let $x_t \in \mathbb{R}^3$ be the regime features at bar t. A K-state Gaussian HMM has latent states $z_t \in \{1, \dots, K\}$, a transition matrix $A_{ij} = P(z_{t+1} = j \mid z_t = i)$, initial probabilities π_0 and emissions $x_t \mid z_t = k \sim \mathcal{N}(m_k, \Sigma_k)$. We use diagonal $\Sigma_k = \mathrm{diag}(v_k)$. Parameters are fitted by expectation–maximisation. The posterior $\gamma_t(k) = P(z_t = k \mid x_{1:n})$ comes from the forward–backward recursion and the most likely path from Viterbi decoding. The likelihood is invariant to permuting the

state indices, so two fits on overlapping data can return the same regimes under different indices.

B. 2-Wasserstein distance between Gaussians

For probability measures μ, ν on $\mathbb{R}^d$ with finite second moments,

$$W_2^2(\mu,\nu) = \inf_{\gamma\in\Gamma(\mu,\nu)} \int \|x-y\|^2 \, d\gamma(x,y), \tag{1}$$

where $\Gamma(\mu,\nu)$ is the set of couplings with marginals μ and ν. For Gaussians the infimum has the closed form

$$W_2^2\big(\mathcal{N}(m_1,\Sigma_1),\mathcal{N}(m_2,\Sigma_2)\big) = \|m_1-m_2\|^2 + \operatorname{tr}\Big(\Sigma_1+\Sigma_2-2\big(\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2}\big)^{1/2}\Big). \tag{2}$$

With diagonal covariances $\Sigma_i = \operatorname{diag}(s_i^2)$ the matrices commute and the trace term reduces to $\|s_1 - s_2\|^2$, so

$$W_2^2 = \|m_1-m_2\|^2 + \|s_1-s_2\|^2. \tag{3}$$

It compares location and spread on the scale of the features and costs $O(d)$, so no numerical optimal-transport solver is needed.

C. Interventions on a policy input

A trained deterministic policy $a = \pi_\theta(o)$ is a known function of its observation. Write $o = (u, p)$, where p is the regime-probability block. We use $\pi_\theta(u, \mathrm{do}(p{=}p'))$ for the action obtained when p is set to p' and everything else is held fixed. Because π_θ is available in closed form, this intervention needs no causal identification assumptions. It measures the effect of the regime input on the policy's output, not any causal effect in the market. A regime signal is *load-bearing* if such interventions change the action. It is *faithful to its names* if the change has the sign that the names imply.

IV. Proposed Methodology

Fig. 1 shows the pipeline. All code is in the `crta` package of the project repository.

A. Data and causal features

Let C_t be the close of bar t. The features are the log return $r_t = \log(C_t/C_{t-1})$, the 20-bar rolling standard deviation σ_t of r, the trend $\tau_t = C_t/\mathrm{SMA}_{20}(C)_t - 1$, the 14-bar Wilder RSI scaled to $[0,1]$, and the MACD histogram ($\mathrm{EMA}_{12}-\mathrm{EMA}_{26}$ minus its 9-bar signal) divided by C_t. Each feature at t uses rows $\le t$ only. A unit test checks that features computed on a truncated history equal the prefix of the full computation. The regime model uses $x_t = (r_t, \sigma_t, \tau_t)$. The agent sees all five as f_t, rescaled to unit order by a fixed map ϕ ($r, \sigma, \mathrm{MACD}\times 100$; $\tau \times 10$; $\mathrm{RSI}\mapsto 2\,\mathrm{RSI}-1$).

B. Walk-forward HMM refits

The tracker makes a single forward pass over the series (Fig. 2). The first $B = 252$ bars (six weeks) form a burn-in. After that, refit j starts at $t_j = B + jS$ with stride $S = 42$ bars (one week). At each refit a four-state diagonal HMM is fitted by EM (200 iterations, seed 42) to the rows strictly before t_j, standardised with that window's mean $\bar{x}$ and standard deviation s. The model then decodes the window $W_j = \{t_j, \dots, t_j+S-1\}$. Component parameters are mapped back to feature units so that different fits can be compared:

$$\mu_k = \hat{m}_k \odot s + \bar{x}, \qquad \sigma_k = \sqrt{\hat{v}_k} \odot s. \tag{4}$$

C. Template initialisation

The burn-in fit is the only place where names come from moments. The rule ℓ sorts components by mean return. Of the two lowest, the one with lower volatility is *bear* and the other is *crisis*. Of the two highest, the one with lower volatility is *calm* and the other is *bull*. This yields templates $(\bar{\mu}_g, \bar{\sigma}_g)$ for $g \in \mathcal{G} = \{\text{bull}, \text{bear}, \text{calm}, \text{crisis}\}$.

D. Wasserstein identity matching

At refit j the cost of naming component k as regime g is the diagonal Wasserstein distance (3):

$$D_{kg} = \|\mu_k - \bar{\mu}_g\|^2 + \|\sigma_k - \bar{\sigma}_g\|^2, \tag{5}$$

and every component takes its nearest template independently,

$$\pi_j(k) = \arg\min_{g\in\mathcal{G}} D_{kg}. \tag{6}$$

The matching need not be one-to-one: two components may share a template, and a template may receive none. The regime probabilities for $t \in W_j$ are the summed posteriors

$$p_t(g) = \sum_{k:\,\pi_j(k)=g} \gamma_t(k), \tag{7}$$

and an unmatched template has $p_t(g) = 0$.

E. Template smoothing

After window W_j is labelled, every matched template moves a step $\eta = 0.2$ toward its members. Let $w_k = \sum_{t\in W_j}\gamma_t(k)$ and $\omega_k = w_k / \sum_{k'\in M_g} w_{k'}$ for the member set $M_g = \{k : \pi_j(k) = g\}$. Then

$$\bar{\mu}_g \leftarrow (1-\eta)\,\bar{\mu}_g + \eta \sum_{k\in M_g} \omega_k \mu_k, \tag{8}$$

and likewise for $\bar{\sigma}_g$. A template with $M_g = \emptyset$ is left unchanged. The update uses window W_j and affects only windows after it.

F. Hard labels and dwell constraint

The hard label $\hat{h}_t$ is the name of the Viterbi state, $\pi_j(\hat{z}_t)$. Runs shorter than $L = 30$ bars (five days) are merged into the preceding label, with the last issued label carried across window boundaries. Hard labels are used only for reporting and per-regime evaluation. The agent sees the probabilities p_t.

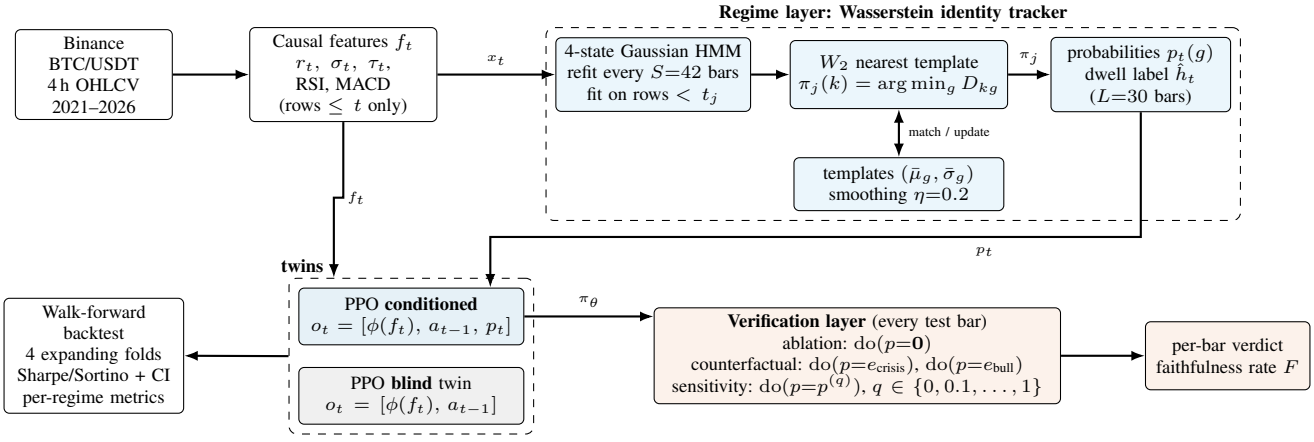


Fig. 1. System architecture. Features are causal. The regime layer refits the HMM on past rows only, names its components by Wasserstein distance to persistent templates, and passes the aggregated probabilities p_t to the conditioned agent. The blind twin differs only in lacking p_t. Every out-of-fold decision of the conditioned agent is audited by three interventions on p_t.

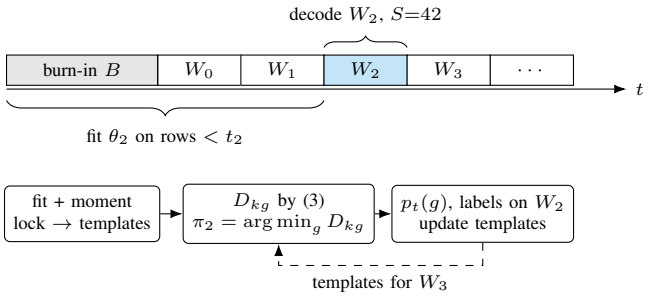


Fig. 2. Causal refit schedule. The burn-in fit names the templates once by moment sorting. Every later window is decoded by a model fitted only on earlier rows, and its components inherit names by Wasserstein matching. Labels already issued are never revised.

G. Twin agents

The trading environment is a Gymnasium episode over the training rows. At bar t the agent chooses a target position $a_t \in [-1, 1]$ and receives

$$R_t = a_t\, r_{t+1} - c\,|a_t - a_{t-1}|, \qquad c = 10^{-3}, \tag{9}$$

that is, the next log return less a 10-basis-point turnover cost. The conditioned agent observes $o_t = [\phi(f_t), a_{t-1}, p_t] \in \mathbb{R}^{10}$ and the blind twin observes $[\phi(f_t), a_{t-1}] \in \mathbb{R}^6$. Both are PPO agents from Stable-Baselines3 with identical settings (Table II). The only architectural difference is four extra input weights in the first layer, and that difference is the experiment.

H. Verification suite

For each out-of-fold bar we freeze the conditioned policy, the market features and the previous position, and intervene on p alone. Let $a_t = \pi_\theta(u_t, p_t)$ be the realised action, e_g the one-hot vector of regime g, and ρ a defensive rank with $\rho(\text{bull}) = 0 < \rho(\text{calm}) = 1 < \rho(\text{bear}) = 2 < \rho(\text{crisis}) = 3$.

Ablation. The regime input is load-bearing at t if

$$\Delta_t^{\text{ab}} = \left|a_t - \pi_\theta(u_t, \mathrm{do}(p{=}\mathbf{0}))\right| \ \ge\ \delta_{\text{ab}} = 0.15. \tag{10}$$

Counterfactual. With $g_t = \arg\max_g p_t(g)$, forcing crisis must cut the position and forcing bull must raise it, each by at least $\delta_{\text{cf}} = 0.10$, whenever the forced regime differs in rank from g_t:

$$\rho(g_t) < 3 \Rightarrow a_t - \pi_\theta(u_t, \mathrm{do}(p{=}e_{\text{crisis}})) \ge \delta_{\text{cf}}, \tag{11}$$

$$\rho(g_t) > 0 \Rightarrow \pi_\theta(u_t, \mathrm{do}(p{=}e_{\text{bull}})) - a_t \ge \delta_{\text{cf}}. \tag{12}$$

Sensitivity. For $q \in \{0, 0.1, \ldots, 1\}$ we inject a crisis probability q and rescale the other entries:

$$p^{(q)} = (1 - q)\,\frac{p_t^{\setminus c}}{\|p_t^{\setminus c}\|_1} + q\, e_{\text{crisis}}, \tag{13}$$

where $p_t^{\setminus c}$ is p_t with its crisis entry set to zero (uniform over the other three if that leaves nothing). The curve $y_i = \pi_\theta(u_t, \mathrm{do}(p{=}p^{(q_i)}))$ passes if it rises at most once, $\sum_i \mathbb{1}[y_{i+1} > y_i + 10^{-8}] \le 1$.

A bar is *faithful* when all three checks pass. The faithfulness rate F is the share of faithful bars. Each bar costs 15 batched forward passes.

I. Live paper-trading console

The repository also includes an interactive console, built with Streamlit, in which both twins trade BTC/USDT with paper money as Binance candles close (Fig. 3). It never sends an order. The user chooses 15-min, 1-h or 4-h candles, as on a charting terminal, and every time stamp is shown in Indian Standard Time (IST, UTC+5:30).

Market feed. Candles come from Binance's public REST API through `ccxt`, which needs no key. A candle that opens at t_0 with width w is kept only once $t_0 + w \le t_{\text{now}}$, so a candle that is still forming never reaches an agent. The 4-h store used for training is updated incrementally. The console requests candles from the last stored open time, lets incoming rows replace overlapping ones, and writes the merged file to a temporary path before renaming it over the old one. If the request fails, the saved file is left untouched and the tape is

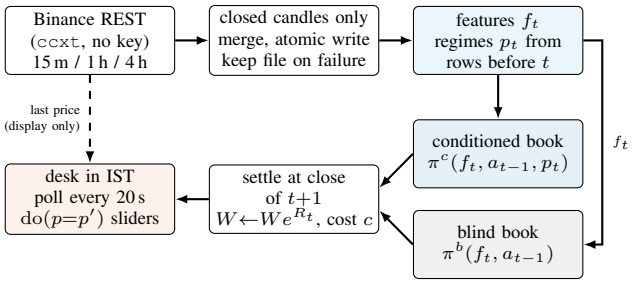


Fig. 3. Live paper-trading console. Only closed candles reach the agents. Both books trade the same tape and are settled with the training reward (9). The last traded price is shown but never used for a decision.

marked stale. The 15-min and 1-h tapes keep their own caches of at most twice the display window (1,000 candles, or 500 for 4 h), so a faster timeframe cannot overwrite the training data. Every 20 s the console polls for newly closed candles and for the last traded price. The price is only displayed, and every decision uses closed candles.

Labels. When a tape is opened, one HMM is fitted on its first $\min(80, \max(20, \lfloor n/5 \rfloor))$ rows, its states are named by the moment rule, and the rest of the window is decoded in one pass. Each candle that closes later is decoded by a model fitted on all rows before it, with the last issued label carried across. This is faster than the full tracker but not identical to it, so Section VI-E re-scores the live period with the tracker itself.

Paper books. Both agents start flat with unit wealth on the same tape. At each closed candle t each agent chooses a_t from its own observation. When candle $t{+}1$ closes, its wealth is multiplied by e^{R_t}, with R_t from (9) including the 10-bp turnover cost. The desk plots both wealth curves, each agent's latest fills and the regime vector fed to the conditioned agent. Sliders let the user override the four regime probabilities for the open candle (rescaled to sum to one), and optionally its features and held position. This applies $\mathrm{do}(p{=}p')$ to the conditioned agent alone and shows its bet next to the blind twin's, so the verifier's interventions can be explored by hand.

V. Experimental Setup

A. Data and folds

We use Binance BTC/USDT four-hour candles from 1 Jan 2021 to 22 Sep 2026 (12,544 bars). Feature warm-up leaves 12,510 bars. Four expanding walk-forward folds start with 1,080 training bars, and each test block has about 2,857 bars (Table I). Each test block is scored by agents trained only on the bars before it. The concatenated test blocks give 11,430 out-of-fold bars from 5 Jul 2021 to 22 Sep 2026. Regime labels for every bar come from the single causal tracker pass described above.

B. Implementation details

Table II lists the settings. PPO uses the Stable-Baselines3 defaults for clipping (0.2) and epochs per update (10). Each agent trains for $\max(8192, 10\,N_{\text{train}})$ steps, which is ten passes over its training window: 10,800, 39,370, 67,940 and 96,510 steps for the four folds. HMMs are fitted with `hmmlearn`. If EM leaves a state with an all-zero transition row, that row is reset to uniform before decoding. The full pipeline (292 tracker refits, eight PPO trainings and the audit) ran in 6.6 minutes on a laptop CPU. All 32 unit tests of the repository pass. They include tests for feature leakage, the closed-form distance and label stability under a component swap. Others check that later bars never change a closed window and that a failed Binance request leaves the saved candles untouched. Further tests check that a refresh keeps older rows and replaces overlapping ones, and that the faster timeframes never write to the training cache. The last three cover the native-twin recipe: a windowed episode is one contiguous stretch of the series, no window replays the whole series, and the reward scale multiplies return and cost alike.

TABLE I
Walk-forward folds and per-fold Sharpe ratios (annualised, out of fold)

Fold	Test period	Train	Cond.	Blind	B&H	Cond.†
0	07/2021–10/2022	1,080	−1.95	−1.84	−0.65	−2.32
1	10/2022–02/2024	3,937	0.24	−3.29	1.69	−0.53
2	02/2024–06/2025	6,794	0.53	0.66	1.10	2.24
3	06/2025–09/2026	9,651	0.50	−0.79	−0.39	1.93

†Archived run with per-fold moment-sorted labels (see Sec. V-C).

TABLE II
Implementation settings (from the project code)

Component	Setting
HMM	4 states, diagonal cov., EM 200 it., min. cov. 10^{-3}
Tracker	burn-in $B{=}252$, stride $S{=}42$, $\eta{=}0.2$, dwell $L{=}30$
Policy net	MLP 64−64, tanh, Gaussian action head
PPO	lr $3{\times}10^{-4}$, $\gamma{=}0.99$, $\lambda_{\text{GAE}}{=}0.95$, entropy 0
Rollout	$n_{\text{steps}}{=}256$, batch 64, seed 42, CPU
Cost	$c = 10^{-3}$ per unit turnover
Verifier	$\delta_{\text{ab}}{=}0.15$, $\delta_{\text{cf}}{=}0.10$, 11-point sweep, $\leq$1 uptick
Bootstrap	1,000 i.i.d. resamples, 95% percentile CI

C. Baselines

Blind twin. The same PPO agent without p_t (Sec. IV-G). *Buy-and-hold.* A constant position $a_t = 1$. *Archived moment-sort run.* The repository keeps the outputs of the pipeline as it stood before the tracker was introduced. That version fitted one HMM per fold and named states with ℓ alone, and everything else was identical. Its blind-twin and buy-and-hold results match ours exactly, so the labels are the only change. *Moment sorting on the same refits.* To isolate the identity rule, we also re-name every one of the 292 refits with ℓ instead of (6). This variant is used for label statistics only. *Reference policies for the verifier.* The rule $a = \mathrm{clip}(p^\top w)$ with $w = (1, -0.4, 0.5, -1)$ over (bull, bear, calm, crisis) is faithful by construction. A constant $a = 0.5$ ignores the regime input, and $a = \mathrm{clip}(-p^\top w)$ inverts it.

D. Metrics

Strategy log returns follow (9) with the realised positions. We report the annualised Sharpe ratio $\sqrt{2190}\,\bar{R}/\hat{s}_R$ and the Sortino ratio $\sqrt{2190}\,\bar{R}/\sqrt{\overline{\min(R,0)^2}}$, using 2,190 four-hour bars per year. Each comes with a 95% bootstrap interval. We also report maximum drawdown and total return. Every metric is additionally computed within each dwell-smoothed regime. For the tracker we report label shares, switches per year, run lengths, the mean matched W_2^2 cost, and an identity-jump statistic: the W_2^2 distance between the probability-weighted components that carry the same name at consecutive refits.

E. Timeframe-native twins

Data. The 15-min and 1-h tapes get twins of their own, trained on each timeframe's own candles. We add ETH/USDT as a second asset and treat it exactly like BTC/USDT, with its own tracker pass and its own twins. Binance candles from 1 Jan 2021 to the live snapshot give, after feature warm-up, 202,596 bars on 15 min and 50,627 bars on 1 h for each asset, about 16 and 4 times as many as the 4-h series.

Tracker. The tracker keeps the calendar spans of the 4-h study, a six-week burn-in, a weekly refit and a five-day dwell. In bars this is $(B, S, L) = (1{,}008, 168, 120)$ on 1 h and $(4{,}032, 672, 480)$ on 15 min. It runs once over each full series, so the live labels continue those of the training data.

Training recipe. We first trained 1-h BTC twins with the 4-h recipe of Sec. IV-G: one episode over all 41,369 training bars, ten passes, unscaled reward and no entropy bonus. Both collapsed towards buy-and-hold. On the holdout year the conditioned agent held $+1.00$ on every bar, so it matched buy-and-hold exactly (-23.3%), and ablation moved its position on no bar. The blind twin averaged $+0.98$ and lost 35.2%. With per-bar rewards of order 10^{-3} and one long episode over a rising market, a constant long position is an easy optimum. The native twins are therefore trained on random windows. Each episode is a randomly placed stretch of 2,000 consecutive training bars (83 days on 1 h, 21 days on 15 min), and the agent starts it flat. The reward (9) is multiplied by 100 during training only, PPO adds an entropy bonus, and each twin trains for 500,000 steps. All other settings follow Table II. One recipe is used for both assets and both timeframes.

Splits. For validation, twins trained on bars before 22 Sep 2024 16:00 UTC are scored on the following year. For the holdout, twins retrained on bars before 22 Sep 2025 16:00 UTC are scored on the year up to the start of the live window, and the same frozen weights are then scored on the live window. Only the validation stage was used to set the recipe, and the holdout and live stages were run once.

Recipe selection. The criterion was set before any training: in validation, both twins must vary their position (standard deviation clearly above 0.05) and ablation must move the conditioned agent on a clear share of bars. Returns were not part of it. With an entropy coefficient of 0.01 no twin held a constant position, but on both 15-min tapes the regime input barely mattered. Ablation passed on 1.1% (BTC) and 0.8% (ETH) of validation bars, against 26.1 and 7.5% on 1 h, and the ETH 15-min blind twin was almost buy-and-hold (mean position $+1.00$, standard deviation 0.06). We retrained all four combinations once with a coefficient of 0.05. This was worse on every combination. Both 15-min tapes collapsed exactly, with the BTC blind twin and the ETH conditioned agent at $+1.00$ on every bar, and ablation passed on at most 3.4% of bars anywhere. We therefore kept 0.01 (Table VII) and tried no third setting.

Regime grouping. On fast candles the five-day dwell outlasts almost every run of the Viterbi path, so the hard label collapses to a single name over nearly the whole series: calm on BTC 1 h and ETH 15 min, bear on BTC 15 min, and bear on 94.7% of ETH 1-h bars. Per-regime results for the native twins are therefore grouped by the most likely regime $\arg\max_g p_t(g)$, the input the agent actually sees. Regime names are fixed per tape at burn-in, so the same name need not mean the same market state on different tapes.

F. Live evaluation protocol

Window. The last price used by the walk-forward study is the close of the 22 Sep 2026 12:00 UTC candle. The live study scores every candle that opened from 16:00 UTC (21:30 IST) that day until a snapshot taken on 13 Oct 2026 at 10:47 UTC. The last candle of each tape is dropped because its next return is not yet known. This leaves 1,994 decisions on 15-min candles, 497 on 1-h candles and 123 on 4-h candles, over the same 21 calendar days. The candles are frozen with the study script, so the numbers below can be reproduced offline.

Agents. On 4 h we use the fold-3 twins of the main run, which were trained only on bars before 3 Jun 2025. On 15 min and 1 h we use the native twins of Sec. V-E, trained on bars before 22 Sep 2025, for BTC and for ETH. All weights stay fixed. Both twins start flat at the first live candle and carry their positions forward. Buy-and-hold enters at the same candle.

Labels. Labels come from the causal tracker of Sec. IV, run on each timeframe's own candles over the full history from 2021. On 4 h it uses $B{=}252$ and $S{=}42$, so the live labels are exactly those the study's tracker would have issued. On 15 min and 1 h it uses the calendar-scaled spans of Sec. V-E.

Why native twins. An earlier version of this study ran the 4-h twins on the faster tapes. There the same features are much smaller: the median σ_{20} in the window is 0.14% on 15 min and 0.31% on 1 h, against 0.61% on 4 h and 1.04% over the 4-h training data. The transferred conditioned agent turned over 0.115 and 0.104 units per bar and lost most of its gross gain to costs. Those results measured transfer across scales rather than the regime input, and are replaced here by the native twins.

Statistics. We report net and gross (before cost) total return, maximum drawdown, mean position and mean turnover $|a_t - a_{t-1}|$ per bar. Sharpe ratios use 35,040, 8,760 and 2,190 bars per year for 15 min, 1 h and 4 h, with the i.i.d. bootstrap of Table II. The twins trade the same candles, so their comparison is paired. We resample the per-bar gap $R_t^c - R_t^b$ with a circular block bootstrap (2,000 resamples, blocks of $\lceil\sqrt{n}\rceil$ bars, i.e. 45, 23 and 12), which keeps serial dependence. We report

TABLE III
REGIME LABELS AFTER BURN-IN (12,258 BARS): SHARE OF BARS, MEAN 4 H LOG RETURN AND MEAN σ_{20} PER LABEL

	W_2 tracker (ours)			Moment sort per refit		
Label	share	ret (bp)	σ (%)	share	ret (bp)	σ (%)
bull	0.0%	–	–	7.2%	-0.8	1.54
bear	7.7%	-7.7	1.42	34.2%	-3.1	0.93
calm	88.5%	$+1.5$	0.97	44.2%	$+3.7$	1.04
crisis	3.8%	-8.1	2.18	14.4%	-0.6	1.14
switches / yr		5.5			7.5	
median run (bars)		110			210	
mean matched W_2^2 (10^{-4})		1.88			9.13	

the 95% interval of the summed log-return gap and the share of resamples in which the conditioned agent is ahead. Every live decision of the conditioned agent is also audited by the verification suite, using the position it actually held.

VI. RESULTS AND DISCUSSION

A. Regime identity under refits

Fig. 4 shows the price with the tracker's labels, the bear and crisis probabilities, and both labellings. Table III compares them. Moment sorting assigns a different name than the W_2 rule to at least one component in 291 of the 292 refits, and to 49.6% of all components. Its labels are semantically unstable. Over the sample, the bars it calls *bull* have a negative mean return (-0.8 bp), and its *crisis* bars are only slightly more volatile than its *calm* bars (1.14% against 1.04%). The W_2 tracker keeps the ordering the names promise: crisis bars average 2.18% volatility and -8.1 bp per bar, against 0.97% and $+1.5$ bp for calm. It also switches less often (5.5 against 7.5 times per year). Its matched cost is 4.9 times lower (1.88 against 9.13×10^{-4}), and the identity jump of the crisis name falls by a factor of 7.4.

These gains have a price. Matching (6) is non-injective in 291 of the 292 refits. Fig. 5 shows why. The bull template inherits the burn-in's high-volatility, high-return component from early 2021. Later fits never produce a component near it, so it receives members in only 35 refits. Under (8) an unmatched template never moves, so after mid-2022 the bull template stays frozen. Calm absorbs the slack and covers 88.5% of bars, and no bar is labelled bull after burn-in. Fig. 6 shows a refit during the June 2022 crash. Components 0 and 2 both map to calm, while moment sorting would have called component 0 bull. A diagnostic variant (not part of the pipeline) replaces (6) with a one-to-one Hungarian assignment. It brings bull back to 5.1% of bars, with a mean return of $+11.9$ bp, but crisis then grows to 31.3% and its volatility falls to 1.15%. Neither rule alone keeps both coverage and sharp semantics.

B. Trading performance

Table IV and Fig. 7 summarise performance out of fold. With W_2 labels the conditioned agent has a Sharpe ratio of -0.16 and loses 29% over the test period. Its blind twin does much worse (-1.12, -93%), and buy-and-hold gains 156% with a Sharpe of 0.35. All bootstrap intervals for the conditioned agent overlap buy-and-hold, and only the blind twin's interval excludes zero. The archived run with moment-sorted labels reached 0.41, mostly through folds 2 and 3 (Table I). Its interval overlaps ours, so the regime-layer change cannot be said to have hurt returns significantly.

TABLE IV
OUT-OF-FOLD PERFORMANCE, 11,430 BARS (95% BOOTSTRAP CI)

System	Sharpe	Sortino	MDD	Return
Conditioned (W_2)	-0.16 [-0.98, 0.63]	-0.22 [-1.34, 0.91]	-77.0%	-29.3%
Blind twin	-1.12 [-1.96, -0.27]	-1.51 [-2.55, -0.37]	-94.4%	-93.0%
Buy-and-hold	0.35 [-0.51, 1.24]	0.50 [-0.71, 1.81]	-77.0%	$+156.2$%
Cond. (archived)	0.41 [-0.41, 1.20]	0.58 [-0.58, 1.77]	-79.8%	$+139.2$%

TABLE V
PER-REGIME SHARPE RATIO AND MEAN POSITION (OUT OF FOLD, W_2 LABELS; NO BAR IS LABELLED BULL)

		Sharpe			mean position	
Regime	n	Cond.	Blind	B&H	Cond.	Blind
bear	852	-0.19	-3.20	-1.94	-0.23	$+0.17$
calm	10,364	-0.17	-1.01	0.71	$+0.41$	$+0.25$
crisis	214	0.19	1.22	-2.23	-0.17	$+0.24$

The per-regime view (Table V) shows where the gap comes from. In bear and crisis segments the conditioned agent holds a net short position on average (-0.23 and -0.17), while the blind twin stays long ($+0.17$ and $+0.24$). Its Sharpe ratio in bear segments, -0.19, is far better than the blind twin's -3.20 and buy-and-hold's -1.94. In calm segments it is long but trails buy-and-hold. Crisis segments cover only 214 test bars, about five weeks, so their annualised ratios are very noisy. This is consistent with an agent that uses its regime input, at least for regimes it saw in training.

C. Faithfulness

Fig. 8 reports the pass rate of each check. The reference policies behave as designed. The faithful rule passes on 97.9% of bars; it fails ablation only where its output is already within 0.15 of zero. The ignoring and inverted rules pass on no bar.

For the conditioned PPO agent with W_2 labels, ablation passes on 60.7% of bars, with a median $|\Delta^{\text{ab}}|$ of 0.42, so the regime input is clearly load-bearing. The counterfactual check, however, passes on only 0.9% of bars, and the combined faithfulness rate is 0.1% (11 of 11,430 bars). The archived agent reached 33.5%. Fig. 9 explains the failure. Forcing *bear* sends the median position to -1, and the forced-bear position lies below the forced-calm position on 76.5% of bars. The agent has learned a defensive response, but it is keyed to bear. Forcing *crisis* gives a median of $+0.64$, which is defensive relative to calm on only 30.1% of bars. Forcing *bull* gives

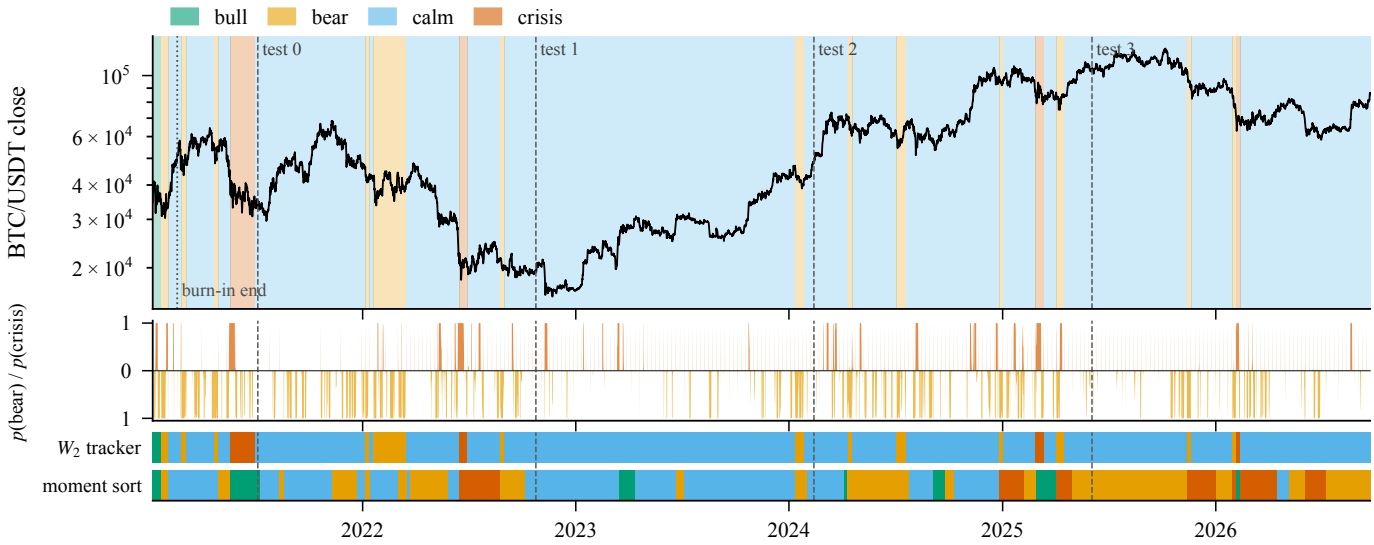


Fig. 4. BTC/USDT four-hour close (log scale) shaded by the dwell-smoothed W_2 tracker label (top). Tracker probabilities of crisis (up) and bear (down) (middle). Labels from the W_2 tracker and from moment sorting on the same 292 refits (bottom). Dashed lines mark the start of each test fold.

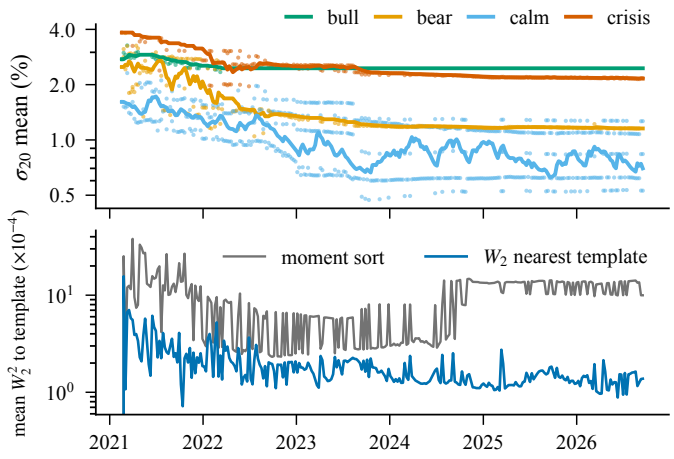


Fig. 5. Top: template volatility (lines) and fitted component volatilities coloured by their W_2 assignment (dots) at each refit. The bull template freezes once it stops receiving members. Bottom: mean squared W_2 distance between components and the template they are named after, for the W_2 rule and for moment sorting.

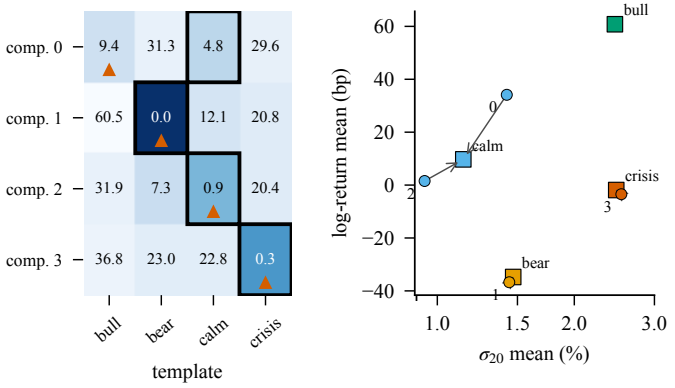


Fig. 6. Refit of 15 Jun 2022. Left: W_2^2 cost D_{kg} ($\times 10^{-4}$). Boxes mark the W_2 assignment and triangles the moment-sort assignment. Right: components (circles) and templates (squares) in mean-volatility and mean-return space, with arrows to the assigned template.

$+0.53$, below the realised median of $+0.74$. Taken separately, the two directions of the counterfactual check pass on 23.3% (crisis cuts the position) and 19.9% (bull raises it) of eligible bars, and their conjunction almost never holds. The crisis sweep in Fig. 9(a) tells the same story. For bear and crisis bars the mean position *rises* as crisis probability is injected.

This behaviour follows from the regime layer. The tracker's probabilities are almost one-hot (mean maximum 0.975), and crisis occupies only 3.8% of bars. One-hot bull never occurs in training after burn-in. The counterfactual inputs e_{bull} and e_{crisis} therefore lie far from, or entirely outside, the data the policy was fitted on. What the agent does there is an extrapolation, so the check fails without the agent being "miswired" on the inputs it actually meets. The archived labels kept all four regimes populated (bull 6.1%, crisis 12.4% of test bars), and their counterfactuals were in distribution. That, more than any quality of the agent, explains the archived 33.5%. Two further caveats concern the checks themselves. The counterfactual rank uses $\arg\max_g p_t(g)$, which agrees with the dwell-smoothed label on only 78.8% of bars. And the sensitivity check is weak, since a flat curve passes it. On 18 Jun 2022, near the crash low, $p_t(\text{crisis}) = 1$ and the agent is fully long. Every intervention returns $+1$, so ablation and counterfactual fail while sensitivity passes. The ignoring policy passes sensitivity on every bar for the same reason.

The lesson is methodological. Swapping one well-motivated regime component for another, with architecture, data and training unchanged, improved label semantics and reversed the faithfulness verdict. A verdict cannot be carried over to

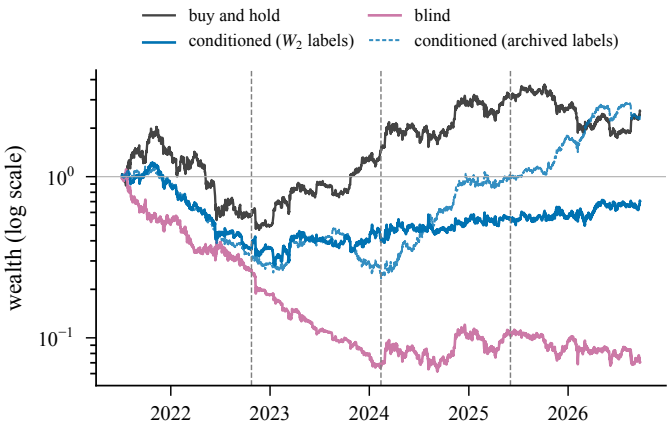


Fig. 7. Out-of-fold wealth (log scale). Dashed vertical lines separate test folds. The dashed blue curve is the archived conditioned agent trained on moment-sorted labels.

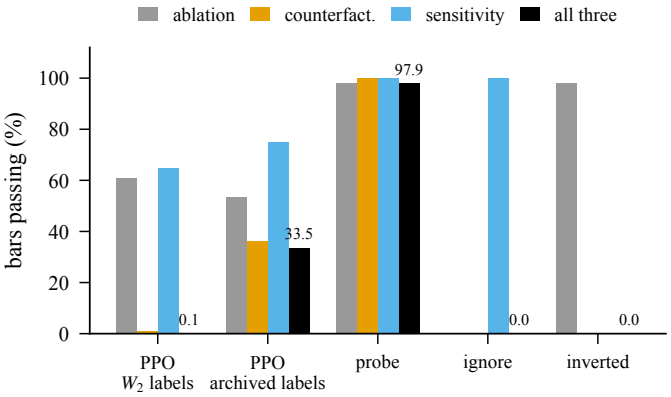


Fig. 8. Share of out-of-fold bars passing each check, for the PPO agent trained on W_2 labels, the archived PPO agent trained on moment-sorted labels, and three reference policies scored on the W_2 probabilities. Numbers give the share passing all three checks.

a modified pipeline.

D. Timeframe-native twins

In validation the selected recipe gave every twin a varying position (standard deviation 0.06 to 0.73), and the conditioned agent finished ahead of its blind twin on all four tapes, with $P(\text{gap}{>}0)$ between 0.58 and 0.95 (Table VII). The holdout does not repeat this. Table VI and Fig. 10 cover 22 Sep 2025 to 22 Sep 2026, a year in which BTC fell by about 23% and ETH by about 35%. Retrained on 27% more bars, including a validation year in which both assets rose by 62 to 81%, most twins drift back towards a fixed position. Three of the eight are essentially constant (position standard deviation at most 0.03): the BTC 15-min conditioned agent is long, the ETH 15-min blind twin is short on every bar, and the ETH 1-h blind twin is long on every bar and therefore identical to buy-and-hold. Two more conditioned agents, BTC 1 h and ETH 15 min, are long on average by $+0.96$ and $+0.97$.

Only the ETH 1-h conditioned agent trades actively and uses its regime input. Ablation moves its position on 27.2% of holdout bars. On bars whose most likely regime is crisis (16.8% of the year) it is short (-0.99 on average) and gains 145.9%, while its blind twin stays long and loses 68.0% there. Over the year it gains 62.6% net and 194.2% gross, against -34.1% for both the blind twin and buy-and-hold, and turnover costs it 59.3 log-return points. Even this gap is not significant: the block-bootstrap interval of $[-42.8, 228.6]$ points contains zero, and the conditioned agent is ahead in 91% of resamples. It also loses 49.4% on bull bars, where it cuts its position to $+0.29$.

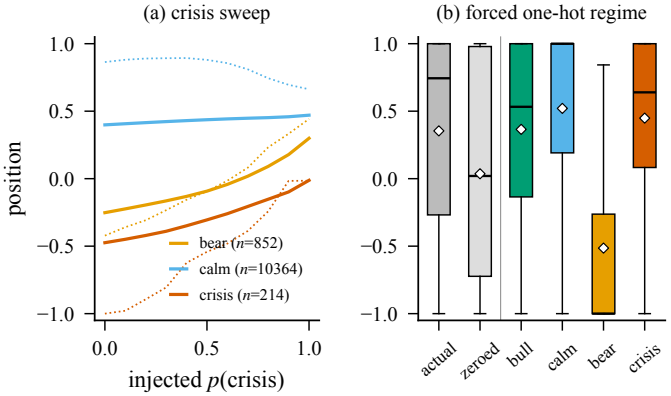


Fig. 9. Interventions on the conditioned agent (W_2 labels). (a) Mean (solid) and median (dotted) position against injected crisis probability, grouped by the bar's label. (b) Distribution of the realised position, the position with the regime input zeroed, and positions under one-hot regime inputs (boxes: interquartile range; diamonds: means).

Elsewhere the conditioned agent does worse than buy-and-hold. On BTC 15 min it is long throughout and trails buy-and-hold by 14.9 points, the only interval in the table that excludes zero ($[-20.7, -9.7]$). Most of that is its own turnover cost (12.4 points). On BTC 1 h it is ahead of its blind twin (29.3 points, $P = 0.74$) but behind buy-and-hold. On ETH 15 min it is long and loses 51.0%, while the blind twin, short on every bar, gains 52.8%. Across the four tapes the conditioned agent leads its twin twice and trails it twice, and no paired interval excludes zero. Where a twin has collapsed, the comparison is decided by the sign of its fixed position in a falling year rather than by the regime input. The faithfulness verdict is the same on all four tapes. Ablation passes on 0.3 to 27.2% of bars and the median ablation effect is zero everywhere. Counterfactual checks pass on at most 6.6% of bars, and at most 2.2% pass all three.

E. Live market comparison

Table VIII and Fig. 11 compare the twins over the live window of Sec. V-F. The 15-min and 1-h rows now come from the native twins of Sec. VI-D and the 4-h rows from the fold-3 twins of the main run. Bitcoin fell by 2.7 to 3.4% across the window, so buy-and-hold lost money on every timeframe. The blind twin finished ahead of the conditioned agent on all three, by 7.6, 8.9 and 10.9 log-return points on 15-min, 1-h and 4-h candles. None of these gaps is significant (Table XI). Every 95% block-bootstrap interval contains zero, and the

TABLE VI

HOLDOUT YEAR OF THE NATIVE TWINS (22 SEP 2025–22 SEP 2026): TOTAL RETURN NET AND GROSS OF COST, ANNUALISED SHARPE RATIO (95% BOOTSTRAP CI), MAXIMUM DRAWDOWN, MEAN AND STANDARD DEVIATION OF THE POSITION, MEAN TURNOVER PER BAR, THE SUMMED LOG-RETURN GAP OF THE CONDITIONED AGENT MINUS THE SYSTEM IN THAT ROW ($\times 100$, 95% CIRCULAR BLOCK-BOOTSTRAP CI) WITH THE SHARE OF RESAMPLES IN WHICH IT IS POSITIVE, AND THE SHARE OF DECISIONS PASSING ABLATION AND ALL THREE CHECKS (%)

System	Net	Gross	Sharpe	MDD	Pos.	s.d.	Turn.	Gap	$P(\text{gap}>0)$	Abl.	All
BTC 15 min: 35,040 bars, BTC -23.6%											
Cond.	-34.2%	-25.5%	-0.93 $[-2.91, 1.08]$	-59.7%	$+1.00$	0.03	0.0035	–	–	0.7	0.7
Blind	-29.7%	-28.6%	-0.78 $[-2.75, 1.20]$	-39.9%	-0.66	0.75	0.0004	-6.7 $[-156.6, 150.3]$	0.45		
B&H	-23.7%	-23.6%	-0.60 $[-2.57, 1.42]$	-53.8%	$+1.00$	–	–	-14.9 $[-20.7, -9.7]$	0.00		
BTC 1 h: 8,760 bars, BTC -23.2%											
Cond.	-29.8%	-19.7%	-0.82 $[-2.92, 1.08]$	-57.5%	$+0.96$	0.28	0.0153	–	–	1.5	1.0
Blind	-47.7%	-37.4%	-1.48 $[-3.41, 0.38]$	-52.4%	$+0.58$	0.81	0.0205	29.3 $[-60.2, 127.1]$	0.74		
B&H	-23.3%	-23.2%	-0.61 $[-2.59, 1.41]$	-53.7%	$+1.00$	–	–	-8.9 $[-47.0, 33.3]$	0.34		
ETH 15 min: 35,040 bars, ETH -34.6%											
Cond.	-51.0%	-49.9%	-1.14 $[-3.10, 0.91]$	-69.6%	$+0.97$	0.22	0.0006	–	–	0.3	0.0
Blind	$+52.8\%$	$+52.9\%$	0.68 $[-1.38, 2.61]$	-45.8%	-1.00	0.00	0.0000	-113.7 $[-334.6, 124.2]$	0.18		
B&H	-34.7%	-34.6%	-0.68 $[-2.61, 1.38]$	-68.0%	$+1.00$	–	–	-28.8 $[-86.7, 14.1]$	0.11		
ETH 1 h: 8,760 bars, ETH -34.1%											
Cond.	$+62.6\%$	$+194.2\%$	0.81 $[-1.12, 2.74]$	-38.4%	$+0.59$	0.80	0.0677	–	–	27.2	2.2
Blind	-34.1%	-34.1%	-0.69 $[-2.73, 1.27]$	-67.9%	$+1.00$	0.00	0.0001	90.3 $[-42.8, 228.6]$	0.91		
B&H	-34.1%	-34.1%	-0.69 $[-2.73, 1.27]$	-67.9%	$+1.00$	–	–	90.3 $[-42.8, 228.6]$	0.91		

TABLE VII

VALIDATION YEAR (22 SEP 2024–22 SEP 2025) OF THE NATIVE TWINS FOR THE TWO ENTROPY COEFFICIENTS TRIED: NET RETURN (%), STANDARD DEVIATION OF THE POSITION, $P(\text{GAP}>0)$ AGAINST THE BLIND TWIN, AND ABLATION AND ALL-CHECK FAITHFULNESS RATES (%)

	net return			pos. s.d.			faithful	
Tape	Cond.	Blind	B&H	Cond.	Blind	P	Abl.	All
entropy 0.01 (used)								
BTC 15 min	$+84.2$	-1.2	$+80.3$	0.46	0.66	0.95	1.1	1.0
BTC 1 h	$+112.0$	$+77.0$	$+79.2$	0.33	0.19	0.93	26.1	0.0
ETH 15 min	$+70.5$	$+27.9$	$+63.3$	0.25	0.06	0.87	0.8	0.0
ETH 1 h	-0.5	-7.1	$+61.8$	0.73	0.55	0.58	7.5	0.8
entropy 0.05								
BTC 15 min	$+78.3$	$+80.3$	$+80.3$	0.20	0.00	0.45	0.0	0.0
BTC 1 h	$+24.6$	$+49.6$	$+79.2$	1.00	0.33	0.38	0.3	0.1
ETH 15 min	$+63.3$	-62.2	$+63.3$	0.00	0.96	0.91	0.0	0.0
ETH 1 h	$+73.5$	-52.9	$+61.8$	0.89	0.54	0.97	3.4	0.2

conditioned agent is ahead in 27, 10 and 5% of resamples. Over three weeks, the annualised Sharpe intervals are more than 14 units wide and say little.

On the faster tapes the gap comes from the positions the native twins carried out of the holdout (Table X). The BTC conditioned agents are long on every live bar (position standard deviation 0.02 on 15 min and 0.01 on 1 h), so they track buy-and-hold to within 0.7 log points. The 15-min blind twin is short on every bar and gains 3.4%. The 1-h blind twin is the only active BTC twin. Its mean position is $+0.43$ with a standard deviation of 0.90, and it gains 5.7%. Native training has removed the over-trading of the transferred 4-h agents, whose conditioned turnover was 0.115 and 0.104 units per bar on these tapes. The native conditioned agents turn over 0.002 and 0.003 units, but only because they hardly trade. On 4 h every live candle is labelled calm, in line with the 88.5% calm share of Table III, so the regime input is almost constant and cannot add information. The conditioned agent stayed mostly long ($+0.82$) and lost a little more than buy-and-hold. The blind twin held about half that exposure and gained 6.2%. Their positions are weakly anti-correlated (-0.21), and the gap reflects how differently the two policies read the same market features.

Ether fell by about 8.5% over the window, more than twice as far as Bitcoin (Table IX, Fig. 12). The ETH twins keep their holdout behaviour. On 15 min the conditioned agent is long and the blind twin short on every bar, so the conditioned agent equals buy-and-hold (-8.8%) and the blind twin gains 9.5%. On 1 h the blind twin equals buy-and-hold (-8.5%), and the conditioned agent is again the only one that responds to its regime input. It is short (-1.00 on average) on bars whose most likely regime is crisis (9.5% of the window) and mostly long elsewhere, and it finishes level with its twin (-8.1

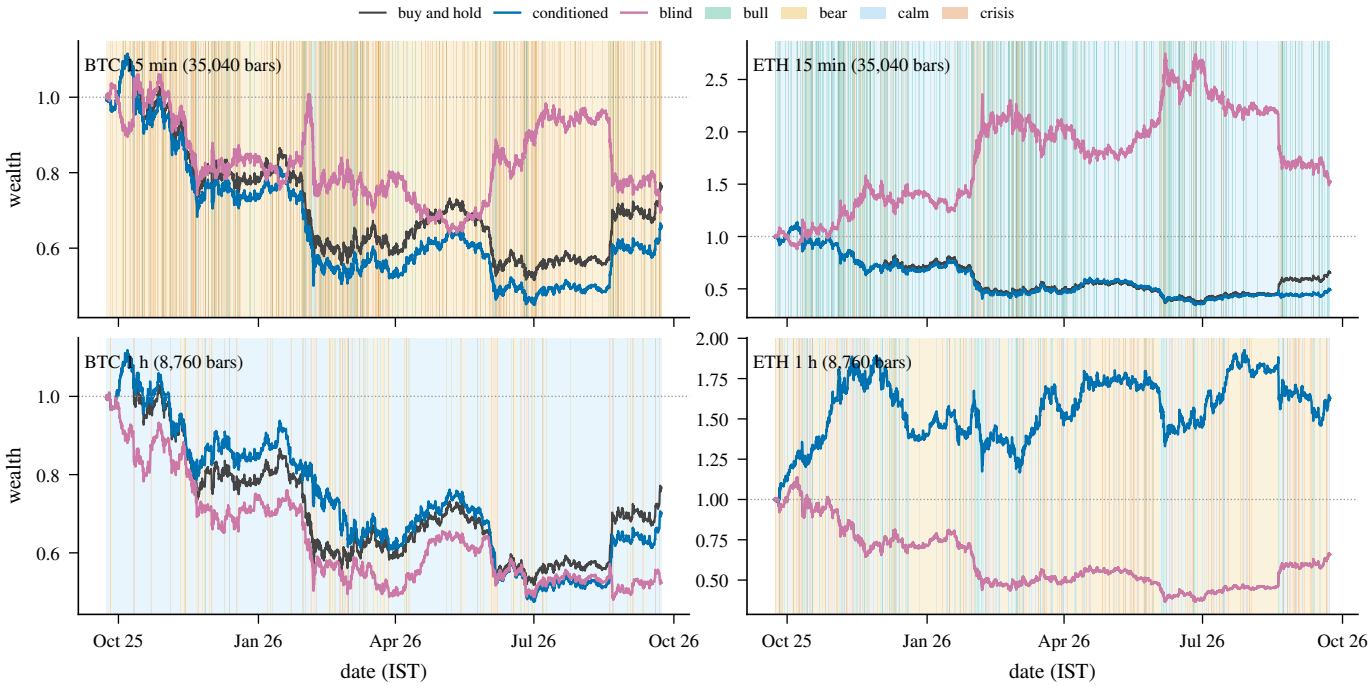


Fig. 10. Holdout-year wealth of the native twins on 15-min and 1-h candles (22 Sep 2025–22 Sep 2026, IST), net of the 10-bp cost. Shading shows the most likely regime $\arg\max_g p_t(g)$. The ETH 1-h blind twin holds $+1$ throughout, so its curve covers buy-and-hold.

TABLE VIII
LIVE PAPER TRADING ON BTC/USDT, 22 SEP–13 OCT 2026: TOTAL RETURN NET AND GROSS OF COST, ANNUALISED SHARPE RATIO (95% BOOTSTRAP CI), MAXIMUM DRAWDOWN, MEAN POSITION AND MEAN TURNOVER PER BAR. 15-MIN AND 1-H TWINS ARE NATIVE, 4-H TWINS ARE THE FOLD-3 AGENTS

System	Net	Gross	Sharpe	MDD	Pos.	Turn.
15 min: 1,994 bars, BTC -3.4%						
Cond.	-4.1%	-3.7%	-2.39 $[-10.43, 5.79]$	-7.9%	$+1.00$	0.002
Blind	$+3.4\%$	$+3.5\%$	1.88 $[-6.46, 9.87]$	-5.0%	-1.00	0.001
B&H	-3.5%	-3.4%	-1.99 $[-10.05, 6.26]$	-7.4%	$+1.00$	–
1 h: 497 bars, BTC -3.2%						
Cond.	-3.3%	-3.2%	-1.87 $[-9.22, 6.47]$	-7.3%	$+1.00$	0.003
Blind	$+5.7\%$	$+6.7\%$	3.11 $[-5.29, 11.49]$	-4.6%	$+0.43$	0.019
B&H	-3.3%	-3.2%	-1.90 $[-9.26, 6.44]$	-7.4%	$+1.00$	–
4 h: 123 bars, BTC -2.7%						
Cond.	-4.8%	-3.5%	-3.08 $[-11.41, 5.33]$	-7.5%	$+0.82$	0.108
Blind	$+6.2\%$	$+7.1\%$	3.73 $[-4.70, 11.97]$	-3.9%	$+0.45$	0.072
B&H	-2.8%	-2.7%	-1.73 $[-9.74, 7.24]$	-6.5%	$+1.00$	–

TABLE IX
LIVE PAPER TRADING ON ETH/USDT WITH NATIVE TWINS, 22 SEP–13 OCT 2026 (COLUMNS AS IN TABLE VIII)

System	Net	Gross	Sharpe	MDD	Pos.	Turn.
15 min: 1,994 bars, ETH -8.7%						
Cond.	-8.8%	-8.7%	-4.21 $[-12.31, 3.59]$	-13.2%	$+1.00$	0.001
Blind	$+9.5\%$	$+9.6\%$	4.11 $[-3.68, 12.30]$	-6.2%	-1.00	0.001
B&H	-8.8%	-8.7%	-4.21 $[-12.31, 3.59]$	-13.2%	$+1.00$	–
1 h: 497 bars, ETH -8.4%						
Cond.	-8.1%	-5.4%	-3.78 $[-11.21, 4.51]$	-9.1%	$+0.79$	0.058
Blind	-8.5%	-8.4%	-3.95 $[-11.64, 4.19]$	-13.1%	$+1.00$	0.002
B&H	-8.5%	-8.4%	-3.95 $[-11.64, 4.19]$	-13.1%	$+1.00$	–

against -8.5%, $P = 0.51$). Its gross return of -5.4% is the best of the three, but turnover costs it 2.9 points. The two assets therefore agree in direction only where both twins have collapsed to fixed signs: the short blind twin wins on both 15-min tapes. Neither asset shows a significant live gap between the twins.

The live audit repeats the holdout verdict (Table XI). Ablation moves the 4-h conditioned agent on 83.7% of live decisions and the ETH 1-h agent on 23.9%, but the native BTC 15-min, BTC 1-h and ETH 15-min agents on at most 0.4%. Their regime input is no longer load-bearing, because a policy that holds $+1$ whatever it sees cannot respond to it. Sensitivity passes on every native decision for the same reason, since a flat curve passes. The counterfactual check passes on at most 1.6% of decisions on any tape, and at most 1.2% pass all three. The native twins thus remove the scale mismatch

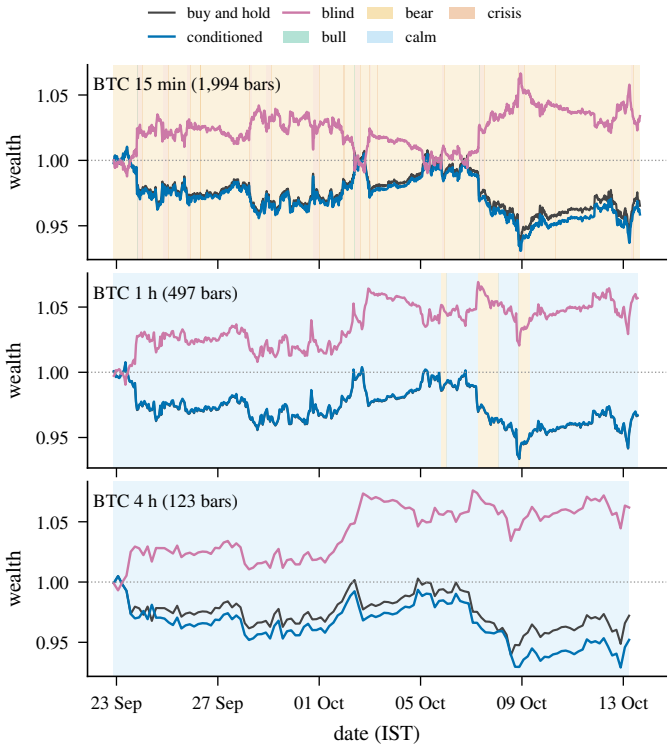


Fig. 11. Live paper-trading wealth on BTC/USDT candles closed after the research data (22 Sep–13 Oct 2026, IST). The 15-min and 1-h twins are native to their timeframe and shading shows $\arg\max_g p_t(g)$. The 4-h twins are the fold-3 agents and shading shows the dwell-smoothed label. All returns are net of the 10-bp cost.

TABLE X
LIVE RESULTS BY REGIME: SHARE OF BARS, MEAN POSITION AND NET RETURN OF EACH TWIN. NATIVE TAPES ARE GROUPED BY $\arg\max_g p_t(g)$, THE 4-H TAPE BY ITS DWELL-SMOOTHED LABEL

			mean position		net return	
Tape	Regime	share	Cond.	Blind	Cond.	Blind
BTC 15 min	bull	0.2%	+1.00	−1.00	−1.5%	+1.5%
	bear	89.7%	+1.00	−1.00	+2.0%	−2.2%
	crisis	10.1%	+0.98	−1.00	−4.6%	+4.2%
BTC 1 h	bear	7.0%	+1.00	+0.60	+0.3%	+0.5%
	calm	93.0%	+1.00	+0.41	−3.5%	+5.2%
BTC 4 h	calm	100%	+0.82	+0.45	−4.8%	+6.2%
ETH 15 min	bull	7.8%	+1.00	−1.00	−0.2%	+0.2%
	calm	91.3%	+1.00	−1.00	−4.8%	+4.8%
	crisis	0.9%	+1.00	−1.00	−4.1%	+4.2%
ETH 1 h	bull	1.0%	+0.54	+1.00	+0.4%	+1.6%
	bear	86.7%	+0.99	+1.00	−9.1%	−8.3%
	calm	2.8%	+0.90	+1.00	+0.0%	+0.0%
	crisis	9.5%	−1.00	+1.00	+0.7%	−1.9%

of the transferred agents but not the faithfulness problem. On most tapes they replace a load-bearing but unfaithful regime input with one that is not used at all.

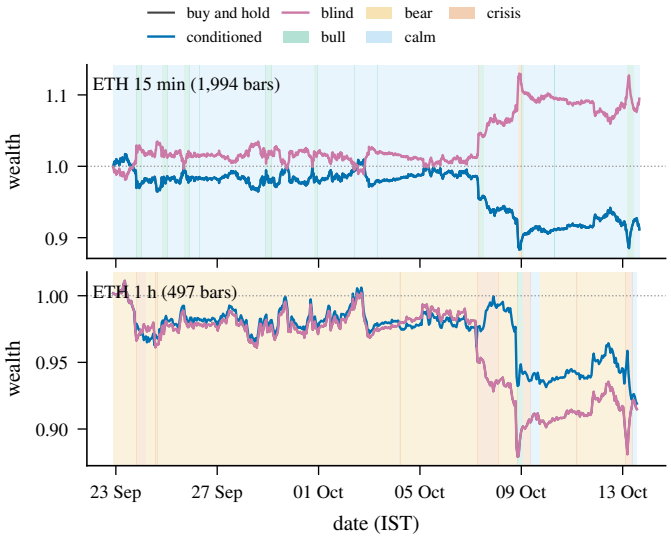


Fig. 12. Live paper-trading wealth of the native ETH/USDT twins (22 Sep–13 Oct 2026, IST). Shading shows $\arg\max_g p_t(g)$. On 15 min the conditioned agent holds $+1$ throughout and its curve covers buy-and-hold; on 1 h the blind twin does.

TABLE XI
LIVE PAIRED COMPARISON AND FAITHFULNESS. GAP: SUMMED LOG RETURN OF THE CONDITIONED AGENT MINUS THE BLIND TWIN ($\times 100$), WITH 95% CIRCULAR BLOCK-BOOTSTRAP CI AND THE SHARE OF RESAMPLES IN WHICH IT IS POSITIVE. RIGHT: SHARE OF LIVE DECISIONS PASSING EACH CHECK (%)

Tape	Gap	$P(\text{gap}>0)$	Abl.	CF	Sens.	All
BTC 15 min	−7.6 [−32.8, 16.5]	0.27	0.4	1.2	100.0	0.4
BTC 1 h	−8.9 [−24.8, 4.5]	0.10	0.4	0.0	100.0	0.0
BTC 4 h	−10.9 [−24.9, 2.5]	0.05	83.7	1.6	91.1	0.0
ETH 15 min	−18.3 [−55.2, 15.9]	0.15	0.3	0.0	100.0	0.0
ETH 1 h	0.5 [−13.1, 16.7]	0.51	23.9	1.6	100.0	1.2

F. Five seeds, after a validation grid

The single-seed native recipe is the control for a one-factor grid. Each cell trains both twins for 500,000 steps on the validation split only (window 2,000, reward scale 100, two threads, seed 42), on BTC and ETH at 1 h and 15 min. The factors, one at a time, are the reward (raw, excess over buy-and-hold, differential Sharpe), the window sampler (uniform, regime-balanced, return-balanced), the entropy coefficient (0.01 or 0.02), a per-episode perturbation of the regime probabilities (0, 0.05, 0.1), and the tracker (v1 or v2). Returns are not a selection input. A tape would pass only if both twins had position standard deviation above 0.05, both spent less than 80% of bars at $|a| > 0.95$, and ablation moved the conditioned agent on at least 10% of bars. The score of a cell is the mean, over the four tapes, of the summed relative shortfalls from those three bars.

No cell passed. Every validation twin still spent more than

90% of bars at $|a| > 0.95$. The baseline had the lowest mean shortfall, 0.985. The next cell, perturbation probability 0.1, was 1.33, and excess-over-buy-and-hold was the worst at 2.65: on BTC 1 h both of those twins finished with position standard deviation 0. Differential Sharpe did raise the standard deviation, often above 0.9, but the mass stayed on the rails and the BTC 15-min blind twin was constant. Return-balanced sampling produced an ablation rate of 73% on BTC 1 h, with standard deviations 0.56 and 0.84, and an extreme-share of 0.98 and 1.00. No factor reduced the baseline shortfall, so the pre-registered stack of helpful factors was empty and no combination was trained. `configs/final.yaml` is the native recipe. The agents below are not regularized.

The same frozen file was then trained five times, seeds 42 through 46, on bars before 22 Sep 2025 16:00 UTC. Holdout and live were scored once each and were not used to change the file. Table XII gives the mean and the standard deviation across seeds. Figure 13 draws each seed faintly and the mean in a heavier line.

Buy-and-hold lost 23.3% and 23.7% on the BTC holdout tapes and 34.1% and 34.7% on the ETH holdout tapes, with no seed variation because the position is fixed. On BTC 1 h holdout the conditioned mean return is ahead of both the blind twin and buy-and-hold, and the conditioned log-return sum is larger on four of the five seeds. The mean Sharpe is still -0.06 (0.86), against -0.72 (0.44) for the blind twin, and the conditioned return moves by 38 percentage points from seed to seed. On ETH 1 h holdout the conditioned mean return is also ahead of the blind twin and of buy-and-hold, but it is ahead on only one seed: the mean is one large path, not a repeated win. On ETH 15 min holdout the blind mean is far ahead (-7.7% against -38.4%). Live, the conditioned agent is ahead of the blind twin on a majority of seeds only for ETH 1 h (four seeds; mean return $+0.4\%$ against -5.5%, and against -8.5% for buy-and-hold). On BTC 1 h live the blind mean is ahead ($+1.8\%$ against 0.0%).

The position standard deviation is not a collapse on every row of Table XII, and it is also not a policy that uses the interior of $[-1, 1]$. The share of bars with $|a| > 0.95$ is between 0.97 and 1.00 for every twin and every window. The ETH 1 h blind twin is flat (standard deviation 0.02 on both windows). The BTC 15-min live blind twin is flat on all five seeds. Where the mean standard deviation is large, the standard deviation of that number across seeds is large too: one seed trades and another sits on a rail. The median absolute ablation effect is 0 on every tape except ETH 1 h live, where the mean of the five medians is 0.27 and their standard deviation is 0.60. Ablation pass rates of 26% and 30% on ETH 1 h are real and still leave the median bar unmoved. Figure 15 compares the legacy verifier with the in-distribution verifier. The second rate is at or below the first. Neither is large. The bull regime was testable on these tapes (no bar was marked untestable). The agents were not trained with the faithfulness penalty, so the check is an independent audit.

On the 1 h tapes the v2 tracker, which was not selected, does change the dwell labels. After burn-in on BTC, v1 issues no bull bar and the dwell label is calm on the whole series. v2 issues bull on 3.7% of post-burn bars and about 0.35 switches per year; bear and crisis dwell runs are still absent, while the argmax of p_t is bull, bear, calm and crisis on 11.6%, 11.3%, 69.7% and 7.4% of those bars. On ETH 1 h, v1 and v2 both leave about 5.4% bull and 94.7% bear. The mean matched W_2^2 on BTC 1 h falls from 4.8×10^{-5} to 2.7×10^{-5}. The forward filter and the online dwell are available, and the trading table above uses v1 because that is what the validation score selected. Agreement between $\arg\max p_t$ and the dwell label on the scored windows is 0.82 (BTC 1 h), 0.74 (BTC 15 min), 0.52 (ETH 1 h) and 0.71 (ETH 15 min).

VII. Limitations and Future Work

Within-window smoothing. Each HMM is fitted only on rows before its window, but decoding uses Viterbi and forward–backward over the whole 42-bar window. The probability at a bar therefore conditions on up to 41 later bars of the same window, and the dwell merge similarly needs a run to finish before it can be judged. Labels are never revised after their window closes, but they are not strictly filtered. A forward-only (filtered) posterior and an online dwell rule would remove this leakage. Until then, the per-regime results should be read with this caveat.

Template starvation. Non-injective matching with a frozen unmatched template removes the bull regime. One-to-one matching is not a complete fix either (Sec. VI-A). Template birth and death, and a longer or multi-cycle burn-in, are natural next steps.

Counterfactual design. One-hot interventions test the policy off its training distribution. Drawing counterfactual regime vectors from the empirical distribution of p_t, and adding a sensitivity criterion that requires a minimum range, would make the verdict more informative.

Statistics. The i.i.d. bootstrap ignores the serial dependence of four-hour returns. A stationary block bootstrap would give wider and more honest intervals. Only the paired gaps of the native and live studies use a block bootstrap. All results come from a single PPO seed, and only the 15-min and 1-h studies include a second asset.

Live evidence. The live study covers 21 days of a falling market. It can show the direction of an effect but cannot establish one. The 15-min and 1-h twins are now trained on their own timeframe, but the random-window recipe did not keep them active. It passed validation, yet after retraining five of the eight holdout twins stay at or near a fixed position, and on those tapes the twin comparison is decided by the sign of that position. The recipe was chosen between only two entropy settings, with one seed. Several seeds, a penalty on constant positions, or a reward measured against buy-and-hold are natural next steps. Paper fills happen at the candle close with a flat 10-bp cost, ignoring slippage, latency and the funding cost of short positions. The console's start-up labelling is faster than the tracker, but it decodes its whole window at once, which smooths within the window over hundreds of candles. Its on-screen wealth curves are therefore illustrative,

TABLE XII

FIVE-SEED NATIVE TWINS AFTER THE FROZEN RECIPE. ENTRIES ARE THE MEAN ACROSS SEEDS 42–46, WITH THE STANDARD DEVIATION OF THOSE FIVE NUMBERS IN PARENTHESES. RETURN IS NET OF TURNOVER. "BEATS" IS THE SHARE OF SEEDS WHOSE CONDITIONED LOG-RETURN SUM EXCEEDS THE BLIND TWIN. ABLATION IS THE LEGACY PASS RATE. THE AGENTS ARE NOT REGULARIZED.

		conditioned		blind			
Tape	Window	return (%)	pos. s.d.	return (%)	pos. s.d.	beats	ablation (%)
BTC 1 h	holdout	2.8 (38.2)	0.55 (0.36)	−26.1 (12.8)	0.44 (0.45)	0.80	7.3 (13.6)
BTC 1 h	live	0.0 (5.6)	0.30 (0.44)	1.8 (4.8)	0.44 (0.45)	0.40	7.5 (14.6)
BTC 15 min	holdout	−27.1 (4.9)	0.09 (0.19)	−25.7 (2.8)	0.15 (0.33)	0.20	0.1 (0.3)
BTC 15 min	live	−1.5 (4.8)	0.17 (0.37)	−2.1 (3.1)	0.00 (0.00)	0.20	0.1 (0.2)
ETH 1 h	holdout	−12.8 (56.4)	0.74 (0.22)	−19.0 (39.6)	0.02 (0.04)	0.20	25.6 (20.0)
ETH 1 h	live	0.4 (9.3)	0.53 (0.14)	−5.5 (8.3)	0.02 (0.05)	0.80	29.5 (30.2)
ETH 15 min	holdout	−38.4 (8.8)	0.36 (0.29)	−7.7 (44.5)	0.37 (0.37)	0.40	0.5 (0.5)
ETH 15 min	live	−5.4 (3.2)	0.33 (0.43)	0.6 (8.9)	0.32 (0.44)	0.40	0.2 (0.2)

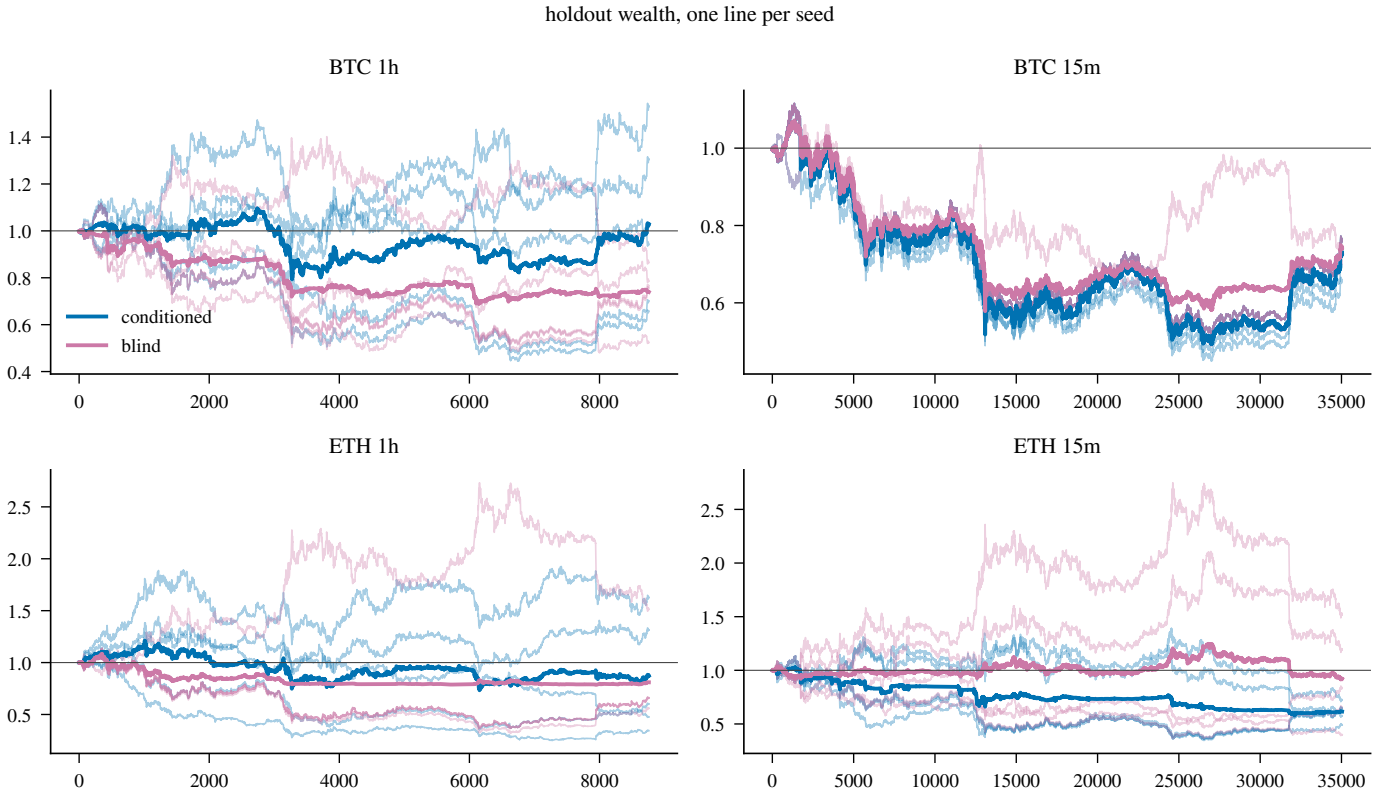


Fig. 13. Holdout wealth for the five frozen seeds. Faint lines are seeds. The heavy line is the mean. Blue is the conditioned twin and pink is the blind twin. The horizontal line is a wealth of 1.

and every live number reported here comes from the study script and the tracker.

Scope. The thresholds δ_{ab} and δ_{cf} and the defensive ordering ρ are design choices. Transaction costs are a flat 10 bp, and the agents were built directly on Stable-Baselines3. Ndoutoumou *et al.* instead place the regime signal inside the FinRL stack [1].

The five-seed section above is the follow-up to the single seed. Tracker v2, the in-distribution counterfactuals, and the differential-Sharpe reward exist and were scored on the validation split. They were not selected. The 4 h study in the earlier sections is still the v1 tracker and the one-hot verifier.

VIII. CONCLUSION

We built a regime-conditioned RL trading agent with a Wasserstein identity tracker and a suite that intervenes on the regime input of every decision. On five years of BTC/USDT four-hour data, the tracker produced labels whose meaning stayed stable across 292 causal refits, which moment sorting did not. The conditioned agent used those labels: removing them changed its position on most bars, and it reduced exposure in bear and crisis segments, where its blind twin did not. Yet it almost never responded to injected regimes in the direction their names imply, because template starvation left the injected states outside its training data. With the same

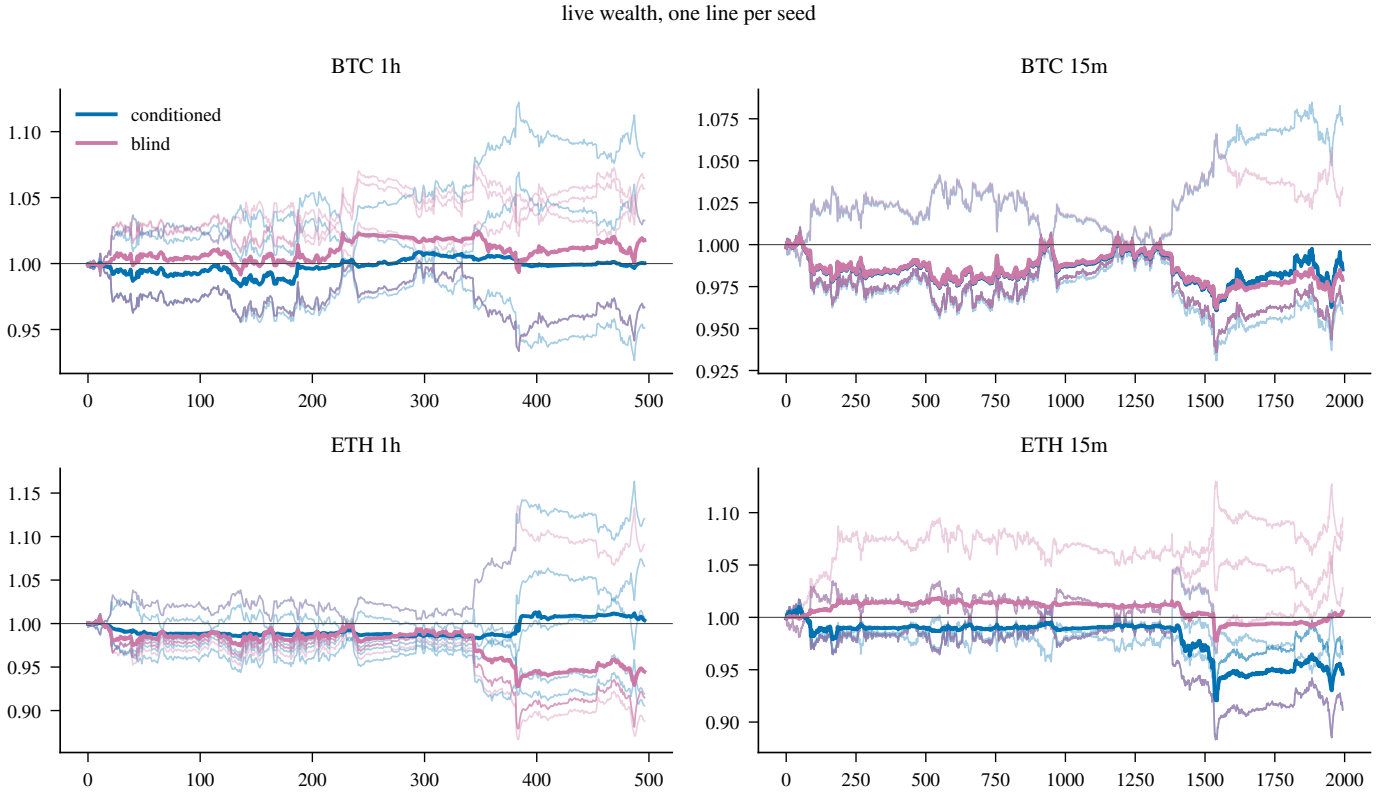


Fig. 14. Live-window wealth for the same five seeds. The scale is the three weeks after 22 Sep 2026, not the holdout year. Faint lines are seeds and the heavy line is the mean.

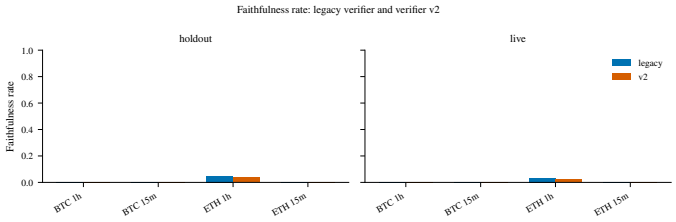


Fig. 15. Mean faithfulness rate across the five seeds. Blue is the legacy one-hot verifier. Orange is verifier v2, which uses training-set probability vectors and rejects a flat sensitivity curve. The axis runs to 1. The bars do not.

agent and data, an earlier labelling gave a faithfulness rate of 33.5%; with the new one the rate is 0.1%. A live paper-trading console and a three-week study on 15-min, 1-h and 4-h Binance candles carried the comparison past the research data. For the faster candles we trained twins on each timeframe's own BTC/USDT and ETH/USDT history. Random-window episodes avoided the buy-and-hold collapse of the single-episode recipe in validation, but not after retraining. On the holdout year five of the eight twins stayed at or near a fixed position, and only the 1-h ETH conditioned agent used its regime input, going short in crisis and finishing ahead of its twin by a margin that was not significant. Live, the blind twin finished ahead on all three BTC timeframes and on 15-min ETH, and no gap between the twins was significant. Where the regime input was load-bearing it was again not faithful, and where a twin collapsed it was not used at all. Regime-aware agents should report a faithfulness verdict next to their returns and recompute it whenever an upstream component changes; the suite here is cheap enough to do so routinely.

A validation grid that was not allowed to look at returns, holdout, or the live window did not find a recipe that keeps both twins off the rails and still uses the regime input. The native recipe remained the least bad of the one-factor cells, and five seeds of that recipe do not reverse the single-seed story. The conditioned twin beats its blind twin on a majority of seeds in two of the eight tape-and-window cells (BTC 1 h holdout, ETH 1 h live) and loses the majority in the other six. Mean Sharpes are negative on the holdout for both twins. The share of time at $|a| > 0.95$ stays above 97%. That is the result.

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